

Exact denominators for M^{-1} : an orbit–stabiliser refinement, and the absorption of $M^{(L)}$ into Macdonald I.6

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8 October 2026 (cycle 2)

Abstract

Let $M_{\rho\nu} = \langle p_\rho, h_\nu \rangle$ on partitions of n . This session had two jobs: a novelty gate (is M the classical m -to- p transition matrix?) and a target (the l.c.m. of the denominators of row ν of M^{-1} is exactly $d_\nu = \prod_i m_i(\nu)!$). Both resolved against the brief. The gate answers *equal*, and not by computation: $\langle h_\nu, m_\sigma \rangle = \delta_{\nu\sigma}$ gives $\langle p_\rho, h_\nu \rangle = [m_\nu]p_\rho$ outright, so M is Macdonald’s matrix $L = M(p, m)$ of I.6 (6.8). The target was already proved in my own paper of the previous cycle; worse, essentially the whole surrounding cluster — the diagonal-block structure, the diagonal entry d_ρ , the determinant, the integral factorisation, and the statement I had planned to prove today as a strengthening — is Macdonald I.6, (6.9)–(6.10) with the Remark on p. 103 and Example 10 on p. 110, whose last sentence is verbatim the strengthening. I record the absorption node by node. I also correct the brief’s assignment of difficulty: it calls “den $| d_\nu$ ” the easy half, by Cramer, but Cramer bounds denominators by $\det M^{(L)} = \prod_\rho d_\rho$, which is not d_ν and is 96 already at $n = 4$ where $d_{(4)} = 1$; the divisibility half is the substantive one and the sharpness half is one line, because the diagonal entry lies in the row.

What survives is a refinement Macdonald’s argument does not reach. He obtains $d_\mu | L_{\lambda\mu}$ from an action of $\text{Aut}(\mu)$ on a set of maps that is *free*. On the inverse side the same group acts on the set partitions $\Pi_{\ell(\lambda)}$ *non-freely*, and the fixed points are exactly where the denominators live. This gives $(M^{-1})_{\lambda\nu} = \sum_O \mu_O / |\text{Stab}(O)|$ over $\text{Aut}(\lambda)$ -orbits in the fibre, hence an *exact* value for every entry whose fibre is a single orbit (351 of 396 support entries for $n \leq 8$) and an upper bound otherwise, tight on 39 of the 45 genuinely falsifiable instances. Two corollaries come out closed: the diagonal entry is exactly $1/d_\lambda$, and the first-column entry is exactly $(-1)^{k-1}(k-1)!/d_\lambda$ with denominator $d_\lambda / \gcd(d_\lambda, (k-1)!)$ — a *proper* divisor of d_λ in 38 of 66 rows. So the exact-denominator theorem, despite reading like a statement about a row, is carried by single entries, and it is not an entrywise statement at all: 267 of 396 support entries have a denominator strictly below d_λ .

1 The object, and the gate

Work in Λ_n over $\mathbb{Q}$ with the Hall inner product, so that $\langle p_\rho, p_\sigma \rangle = \delta_{\rho\sigma} z_\rho$ and $\langle h_\nu, m_\sigma \rangle = \delta_{\nu\sigma}$. For a partition ρ write $\ell(\rho)$ for the number of parts, $m_i(\rho)$ for the multiplicity of i , and

$$d_\rho := \prod_{i \geq 1} m_i(\rho)! = \frac{z_\rho}{\rho_1 \cdots \rho_{\ell(\rho)}} = |\text{Aut}(\rho)|,$$

where $\text{Aut}(\rho) \leq S_{\ell(\rho)}$ is the group of permutations of the *positions* of ρ fixing the sequence of parts. $M = M(n)$ is the matrix $M_{\rho\nu} = \langle p_\rho, h_\nu \rangle$, and $M^{(L)}$ its leading block on $\{\rho : \ell(\rho) \leq L\}$.

The gate asked whether $M^{(L)}$ is a truncation of the classical m -to- p transition matrix. It is, and the brief’s prescribed entrywise comparison was not the right instrument, because the two sides are equal by a one-line duality rather than by a numerical coincidence.

Lemma 1. $M_{\rho\nu} = [m_\nu] p_\rho$ for all $\rho, \nu \vdash n$. Hence M is exactly the matrix $L = M(p, m)$ of Macdonald I.6 (6.8), and $M^{(L)}$ is its leading block for the length filtration.

Proof. Write $p_\rho = \sum_\sigma c_\sigma m_\sigma$. Pairing with h_ν and using $\langle h_\nu, m_\sigma \rangle = \delta_{\nu\sigma}$ gives $\langle p_\rho, h_\nu \rangle = c_\nu$. The bases (h_ν) and (m_σ) are dual, so no computation intervenes. $\square$

Remark 2. This is Lemma 1 of my own previous-cycle paper, where it was recorded as one of “three faces” of the entry. The gate was therefore already answered in the document the gate was written to test, and the entrywise sweep I ran (§5, 204 falsifiable entries, 0 mismatches, two mechanism-disjoint engines) corroborates a lemma rather than deciding a question. I report it as corroboration and not as the decision.

2 What Macdonald I.6 already contains

I hold Macdonald’s book on disk and read the relevant pages first-hand this session: I.6 (6.8)–(6.11) and the Remark on p. 103, and Example 10 on p. 110.

Provenance, because this scan has a known hazard. My own source entry for this book records that its OCR text layer is badly mangled and that “a text-layer grep on this file yields garbage that looks like content”, with the instruction to transcribe from rendered page images. I first read Example 10 off the text layer, in violation of that instruction, and then rendered PDF pages 114 and 121 (book pp. 103 and 110) at 200dpi and read the images directly. Both load-bearing quotations below were confirmed against the rendered images, and the text layer happened to agree on the words that matter. Had it not, the headline of this session would have been noise. Macdonald writes $u_\mu = \prod_{i \geq 1} m_i(\mu)!$ — my d_μ — and $U = \text{diag}(u_\mu)$. The correspondence with the nodes of my registry subtree **what-is-M-L** is as follows.

my node	statement	Macdonald I.6
lem-three-faces	$L_{\lambda\mu} = \#\{f : f(\lambda) = \mu\}$	(6.9)
thm-structure-diagonal-blocks	diagonal blocks are diagonal diagonal entry $L_{\mu\mu} = u_\mu$	(6.9)+(6.10)+Remark p.103 Ex. 10, parenthetical
cor-det-M-L	$\det L = \prod_\mu u_\mu$	two lines from Ex. 10
thm-inverse-mobius	L^{-1} via μ_{Π_k}	Doubilet; already flagged classical
cor-exact-denominators	row ν has l.c.m. exactly d_ν	two lines from Ex. 10
today’s intended target	$(\tilde{m}_\mu)$ is a $\mathbb{Z}$ -basis of $\mathbb{Z}[p]$	Ex. 10, final sentence

Three of these need spelling out.

The diagonal-block structure. Macdonald proves $L_{\lambda\mu} = 0$ unless λ refines μ ($\lambda \leq_R \mu$), and then remarks, p. 103: “two distinct partitions λ and μ of n such that $\ell(\lambda) = \ell(\mu)$ are always incomparable for the relation $\leq_R$.” Distinct same-length partitions are therefore outside the support, i.e. each diagonal block of the length filtration is a diagonal matrix. That was billed as “THE RESULT” of my previous cycle; it is this Remark.

The integral factorisation. Example 10 states that LU^{-1} is a strictly lower unitriangular matrix of non-negative integers, and proves it by letting $S_\mu = \text{Aut}(\mu)$ act on $E_{\lambda\mu} = \{f : [\ell(\lambda)] \rightarrow [\ell(\mu)] : f(\lambda) = \mu\}$ by post-composition. Since every $f \in E_{\lambda\mu}$ is surjective (all $\mu_j > 0$), the action is free, so $u_\mu = |S_\mu|$ divides $|E_{\lambda\mu}| = L_{\lambda\mu}$; and $E_{\mu\mu} = S_\mu$ gives $L_{\mu\mu} = u_\mu$. This is my factorisation $M = \tilde{M}D$ together with the diagonal entry, by my argument, in his parenthesis.

Today's intended target. I had planned to strengthen `cor-exact-denominators` to the basis-independent statement that the lattice $P_n = \mathbb{Z}\text{-span}\{p_\rho\}$ equals $\mathbb{Z}\text{-span}\{\tilde{m}_\nu\}$, where $\tilde{m}_\nu = d_\nu m_\nu$ is the augmented monomial, giving $\Lambda_n^\mathbb{Z}/P_n \cong \bigoplus_{\rho \vdash n} \mathbb{Z}/d_\rho$ and hence the conclusion that no choice of integral bases on either side can reduce the denominators below d_ρ . Example 10's final sentence is: "Since LU^{-1} is unitriangular, $(\tilde{m}_\mu)$ is a $\mathbb{Z}$ -basis of the subring $\mathbb{Z}[p_1, p_2, \dots]$ of Λ ." That is the statement, graded, and the cokernel is one line from it ($\tilde{m}_\nu = d_\nu m_\nu$ is already diagonal). I verified my version computationally before reading the page (§5: $\text{SNF}(M^{(L)}) = \text{SNF}(\text{diag}(d_\rho))$ for all 36 pairs (n, L) with $n \leq 8$, 20 of them with non-cyclic cokernel, 0 mismatches). The verification stands; the novelty does not.

Remark 3 (the standing lesson, now at five). Four absorptions in the preceding 48 hours were of *closed forms* — coordinates on a densely sampled orbit. This one is different in kind and worse: it is a structure theorem, absorbed by a 1995 textbook that was already in `projects/library/` and whose relevant chapter the gate itself named. The gate was pointed at the right page and I ran a numerical comparison instead of turning to it. A novelty gate that prescribes a computation when the source is on the shelf tests the wrong thing.

3 The brief's difficulty assignment, corrected

The brief frames the target as a matched pair of bounds: "Divides d_ν ' is the easy half and is a consequence of Cramer plus det. 'Is not a proper divisor' is the content." Both halves are assigned backwards.

Proposition 4. *Cramer's rule does not yield $\text{den}((M^{(L)})_{\nu\bullet}^{-1}) \mid d_\nu$.*

Proof. Cramer gives entries of $(M^{(L)})^{-1}$ as integers divided by $\det M^{(L)}$, so it bounds denominators by $\det M^{(L)} = \prod_{\ell(\rho) \leq L} d_\rho$, a quantity depending on all of n and L and not on ν . At $n = 4$, $L = 4$ this is 96, while the claim for $\nu = (4)$ is that the denominator is $d_{(4)} = 1$. A bound of 96 does not imply a bound of 1. $\square$

The divisibility half is exactly the substantive content, and it is what Macdonald's Example 10 proves (equivalently what Doubilet's Möbius inversion proves): row λ of M^{-1} is $1/d_\lambda$ times an integer vector, because $M^{-1} = U^{-1}(LU^{-1})^{-1}$ and $(LU^{-1})^{-1}$ is integral unitriangular. The sharpness half is then immediate and needs no new idea.

Corollary 5. *For every ν with $\ell(\nu) \leq L$, the l.c.m. of the denominators of row ν of $(M^{(L)})^{-1}$ is exactly d_ν .*

Proof. Divisibility is the previous paragraph. For sharpness, M is triangular for $\leq_R$ with $M_{\nu\nu} = d_\nu$, and the leading block for an initial segment of a filtration inverts to the inverse of the leading block, so $((M^{(L)})^{-1})_{\nu\nu} = 1/d_\nu$. That entry lies in row ν , so the l.c.m. is divisible by d_ν . $\square$

Remark 6. The asymmetry the brief wanted — $c \mapsto Y$ integral, $Y \mapsto c$ costing exactly $|\text{Aut}(\nu)|$ — is correct and is Corollary 5. What is not correct is that its sharpness is the hard half. The sharpness is carried by one entry, the diagonal one; it cannot be otherwise, since a row l.c.m. is attained somewhere and the diagonal is the only entry whose value the structure theorem pins down for free.

4 What survives: entrywise denominators and a non-free action

Macdonald’s divisibility argument works because the action on $E_{\lambda\mu}$ is free. The question the brief should have asked — the one that quantifies over every entry rather than every row — is for the *exact* denominator of each individual entry. There the relevant action is not free, and the failure of freeness is precisely what produces the denominators.

Fix $\lambda \vdash n$ with $\ell(\lambda) = k$. Let Π_k be the lattice of set partitions of $[k]$; for $\pi \in \Pi_k$ let λ^π be the partition of n with parts the block sums $(\sum_{i \in B} \lambda_i)_{B \in \pi}$, and let

$$\mu(\pi) := \mu_{\Pi_k}(\hat{0}, \pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!.$$

By **thm-inverse-mobius** (classical; Doubilet), $(M^{-1})_{\lambda\nu} = d_\lambda^{-1} \sum_{\pi \in F_{\lambda\nu}} \mu(\pi)$ where $F_{\lambda\nu} = \{\pi \in \Pi_k : \lambda^\pi = \nu\}$ is the *fibre*.

Theorem 7 (orbit decomposition of the entry). *$\text{Aut}(\lambda)$ acts on Π_k by $w \cdot \pi = \{w(B) : B \in \pi\}$. This action preserves each fibre $F_{\lambda\nu}$ and preserves μ . Consequently, summing over the $\text{Aut}(\lambda)$ -orbits $O \subseteq F_{\lambda\nu}$ and writing μ_O for the common value of μ on O ,*

$$(M^{-1})_{\lambda\nu} = \sum_{O \subseteq F_{\lambda\nu}} \frac{\mu_O}{|\text{Stab}(O)|}, \quad \text{hence} \quad \text{den}((M^{-1})_{\lambda\nu}) \mid \text{lcm}_{O \subseteq F_{\lambda\nu}} \frac{|\text{Stab}(O)|}{\gcd(|\text{Stab}(O)|, \mu_O)}.$$

Proof. If $w \in \text{Aut}(\lambda)$ then $\lambda_{w(i)} = \lambda_i$ for all i , so each block sum is unchanged and $\lambda^{w \cdot \pi} = \lambda^\pi$: the fibre is stable. Also $|w(B)| = |B|$, so $\mu(w \cdot \pi) = \mu(\pi)$, and μ is constant on orbits. By orbit-stabiliser, $|O| = d_\lambda / |\text{Stab}(\pi)|$ for any $\pi \in O$, so

$$\sum_{\pi \in F_{\lambda\nu}} \mu(\pi) = \sum_O |O| \mu_O = \sum_O \frac{d_\lambda}{|\text{Stab}(O)|} \mu_O,$$

and dividing by d_λ gives the displayed sum. Each summand $\mu_O / |\text{Stab}(O)|$ in lowest terms has denominator $|\text{Stab}(O)| / \gcd(|\text{Stab}(O)|, \mu_O)$, and the denominator of a finite sum of rationals divides the l.c.m. of the denominators of the summands. $\square$

Corollary 8 (single-orbit fibres are exact). *If $F_{\lambda\nu}$ is a single $\text{Aut}(\lambda)$ -orbit, then $\text{den}((M^{-1})_{\lambda\nu}) = |\text{Stab}| / \gcd(|\text{Stab}|, \mu_O)$ exactly, with no inequality.*

Proof. A single summand is already in lowest terms after cancellation. $\square$

For $n \leq 8$, 351 of the 396 support entries have single-orbit fibres, so Corollary 8 is an exact formula on 89% of the support. The two extreme fibres are single orbits for every λ , and give the two closed forms.

Corollary 9 (the diagonal). *$F_{\lambda\lambda} = \{\hat{0}\}$, and $(M^{-1})_{\lambda\lambda} = 1/d_\lambda$ with denominator exactly d_λ .*

Proof. If $\lambda^\pi = \lambda$ then π has $\ell(\lambda) = k$ blocks, so all blocks are singletons and $\pi = \hat{0}$. (Coarsening strictly decreases the number of parts unless $\pi = \hat{0}$.) Then $\text{Stab}(\hat{0}) = \text{Aut}(\lambda)$ has order d_λ and $\mu(\hat{0}) = 1$. $\square$

Corollary 10 (the first column). $F_{\lambda,(n)} = \{\hat{1}\}$, and

$$(M^{-1})_{\lambda,(n)} = \frac{(-1)^{k-1}(k-1)!}{d_\lambda}, \quad \text{den}((M^{-1})_{\lambda,(n)}) = \frac{d_\lambda}{\gcd(d_\lambda, (k-1)!)}.$$

Proof. $\lambda^\pi = (n)$ forces $\pi = \hat{1}$, whose stabiliser is all of $\text{Aut}(\lambda)$ and whose Möbius value is $(-1)^{k-1}(k-1)!$. Apply Corollary 8. $\square$

Example 11. $\lambda = (1^n)$: $k = n$, $d_\lambda = n!$, and the entry is $(-1)^{n-1}(n-1)!/n! = (-1)^{n-1}/n$, whose denominator n is a proper divisor of $n!$ for $n \geq 3$. Since $m_{(1^n)} = e_n$ this recovers the classical $[p_{(n)}]e_n = (-1)^{n-1}/n$.

So the row-level theorem is not an entrywise theorem, and Corollary 10 shows the failure is systematic rather than sporadic: the first column drops below d_λ whenever $\gcd(d_\lambda, (k-1)!) > 1$, which for $n \leq 8$ happens in 38 of the 66 rows. The measured totals are blunter still.

Proposition 12 (measured, not proved). *For $n \leq 8$: of the 396 entries in the support of M^{-1} , only 129 attain d_λ and 267 have a denominator that is a proper divisor of d_λ . Restricting to the 42 rows with $d_\lambda > 1$ (the other 24 rows are vacuous, every denominator being $1 = d_\lambda$), the l.c.m. is attained only at the diagonal in 18 rows and at some further entry in 24.*

I state Proposition 12 as a measurement and claim no proof of it. In particular I had guessed that the diagonal is the unique attaining entry; it is not, and the 24 counterexample rows begin at $n = 2$ with $\lambda = (1, 1)$, where $\nu = (2)$ also attains $d_\lambda = 2$.

5 Computational verification

All arithmetic is exact: Python integers and `fractions.Fraction`, with Smith normal forms from `sympy.matrices.normalforms`. Following the comparison warning carried into this session, no test uses a bare `!=` on symbolic expressions; every comparison here is between `Fractions` or integers. Code: 2026-10-08-c2-exact-denominators-code/ in this repository (`lattice`, `gate`, `snf`, `entrywise`, `entrywise2`, `final`).

Three mechanism-disjoint engines. Engine A counts maps f with prescribed block sums. Engine B expands $\prod_i \sum_v x_v^{\lambda_i}$ honestly in n variables and reads off the coefficient of the sorted-exponent representative, checking m -symmetry as it goes. Engine C computes $\langle p_\rho, h_\nu \rangle = z_\rho [p_\rho] h_\nu$ from $h_m = \sum_{\sigma \vdash m} p_\sigma / z_\sigma$. Engine C never mentions m ; engine B never mentions h or z . B versus C is the gate test.

Falsifiability census, before any verdict. Of the 434 entries with $n \leq 7$, only 204 lie in the support; the other 230 are forced to 0 on both sides by the support theorem, so agreement there is rank one. Reported: 204 falsifiable, 0 mismatches (B vs C); and 434 compared, 0 mismatches (A vs B).

The factorisation and the cokernel. $M = \widetilde{M}D$ entrywise for $n \leq 8$: 918 entries, 0 failures, with 330 nonzero off-diagonal entries of $\widetilde{M}$ — unitriangularity is non-vacuous only where those are. $\widetilde{M}^{-1}$ integral and equal to the Möbius fibre sums: 434 entries, 160 nonzero off-diagonal, 0 mismatches, integrality asserted entrywise. $\text{SNF}(M^{(L)}) = \text{SNF}(\text{diag}(d_\rho))$ for all 36 pairs (n, L) , $n \leq 8$: 25 have nontrivial cokernel and 20 have *non-cyclic* cokernel, which is the subset where the group carries more than the index; 0 mismatches. Smith normal form was checked to be a non-constant function of the off-diagonal part before being used ($\text{diag}(2, 2) \mapsto \text{diag}(2, 2)$ but $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \mapsto \text{diag}(1, 4)$), unlike $\det$, which the previous cycle proved is a function of the diagonal alone once triangularity holds.

Theorem 7, with the falsifiable set counted first. Tightness of the bound is an *identity* when the fibre is a single orbit, so the claim can only fail on multi-orbit fibres. Of 396 support entries with $n \leq 8$, 351 are single-orbit and only **45** are falsifiable. On those 45: 0 violations of the divisibility, 39 tight, 6 loose.

The 6 loose instances, in full. Every one has exactly two orbits, and in every one the bound exceeds the true denominator by a factor of exactly 2:

λ	ν	entry	den / bound
$(2, 2, 1, 1, 1)$	$(3, 2, 2)$	$2/3$	$3 / 6$
$(2, 2, 2, 1, 1)$	$(6, 2)$	$-5/3$	$3 / 6$
$(2, 2, 1, 1, 1, 1)$	$(3, 2, 2, 1)$	$2/3$	$3 / 6$
$(2, 1^6)$	$(6, 2)$	$-2/3$	$3 / 6$
$(2, 1^6)$	$(4, 2, 2)$	$1/4$	$4 / 8$
$(2, 1^6)$	$(3, 3, 2)$	$2/9$	$9 / 18$

Planted control. The bound reads the *stabiliser order*. To check that it is not satisfied by any plausible group-theoretic quantity of the right size, I planted the off-by-duality error $|\text{Stab}(O)| := |O|$ (orbit size in place of stabiliser order) and recomputed. True stabiliser: **0** violations. Planted orbit size: **248** violations. The test is sensitive. Order observed: non-constant dependence established first (the two quantities differ on every orbit of size $\neq \sqrt{d_\lambda}$), then the plant, then the cardinality, then the verdict.

An instrument fault found and fixed. The first run of the orbit computation reported the bound loose in 95 of 204 cases. The cause was in my own generator of $\text{Aut}(\lambda)$: it appended `tuple(cur)` where `cur` was a `dict`, which yields the dict’s *keys*, so every “group element” was the identity and every orbit had size 1. The guard I had written, `assert len(G)==d_of(lam)`, passed throughout: the cardinality was right while the content was wrong. Replacing it with `assert len(set(G))==d_lam` plus an entrywise check that each w really fixes the part sequence caught it. The corrected run gives the figures above. A cardinality check cannot see the content of what it counts.

6 Where the remaining gap is

Theorem 7 is an equality for the entry and an inequality for the denominator, and §5 locates the inequality precisely: it is strict exactly when two or more orbits cancel against each other, which

for $n \leq 8$ happens in 6 of 45 multi-orbit instances, always by a factor of 2, always with exactly two orbits, and always with λ containing a part 2 alongside 1s. I do not have a characterisation of when cross-orbit cancellation occurs, and I am not confident one in closed form would be worth having: by the standing lesson of this week, a closed form is a coordinate and gets absorbed. The honest statement of the gap is:

Gap. Determine $\gcd(d_\lambda, \sum_{\pi \in F_{\lambda\nu}} \mu(\pi))$, equivalently the exact denominator of $(M^{-1})_{\lambda\nu}$, in the multi-orbit case. Theorem 7 reduces this to the cancellation between $\text{Aut}(\lambda)$ -orbits in a single fibre of the coarsening map; the single-orbit case is Corollary 8 and is closed. The uniform factor 2 in all six observed instances is unexplained, and six instances is too few to lean on.

7 What this session changed in the registry

Demotions owed and applied to `two-part-green-polynomials.json`, subtree `what-is-M-L: lem-three-faces`, `thm-structure-diagonal-blocks`, `cor-det-M-L`, `cor-exact-denominators` are classical (Macdonald I.6, as tabulated in §2); the trust grades stand, since the proofs given are proofs, but the novelty claims are withdrawn. `thm-inverse-mobius` was already marked classical and is unchanged. The strengthening planned for this session is withdrawn entirely as Macdonald I.6 Example 10. Added: `thm-orbit-entrywise-denominators` (Theorem 7 and Corollaries 8–10), with the measured Proposition 12 and the gap above recorded as its open residual.